VNR Vignana Jyothi Institute of Engineering

Mechanical Engineering | Course Code: ME201

Mechanics of Solids & Strength of Materials - Comprehensive Syllabus Lec

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Syllabus Focus: ME201 Full Syllabus Comprehensive Lecture Notes Units 1-5

Summary: Complete syllabus lecture notes covering Mechanics of Solids & Strength of Materials.

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Simple Stresses, Strains & Elastic Moduli

" Syllabus Scope: Normal and shear stress, Hookes law, stress-strain curves for ductile and brittle mater

Unit 2: Compound Stresses & 2D Mohr Circle

" Syllabus Scope: Stresses on inclined planes, principal stresses, principal planes, maximum shear stress

ME201: Mechanics of Solids & Strength of Materials - Advanced Topics & Assessment

Academic Track: Mechanical Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Shear Force & Bending Moment in Beams

" Beam support classifications, point loads, uniformly distributed loads UDL, uniformly varying loads UVL, SFD

Unit 4: Flexural Stresses, Shear Stresses & Deflection

" Pure bending theory Euler-Bernoulli, section modulus, shear stress distribution across I and T sections, dou

Unit 5: Torsion of Circular Shafts & Column Buckling

" Torsion formula for solid and hollow shafts, polar modulus, angle of twist, Eulers column buckling theory, R

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

1. Draw the Shear Force Diagram SFD and Bending Moment Diagram BMD for an overhanging beam subjected to c
2. Given a 2D stress element with sigma_x = 80 MPa, sigma_y = -40 MPa, and tau_xy = 30 MPa, construct Moh
3. Derive the Euler column buckling load P_cr for a slender column with one end fixed and the other end p

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Ferdinand P. Beer, E. Russell Johnston - Mechanics of Materials McGraw-Hill 7th Ed
* James M. Gere, Barry J. Goodno - Mechanics of Materials Cengage Learning 8th Ed

Authorized NoteVault Educational Document " Verified for Registered University Students